

Institute of Actuaries of India

Subject CM1A – Actuarial Mathematics (Paper A)

November 2019 Examination

INDICATIVE SOLUTION

Introduction

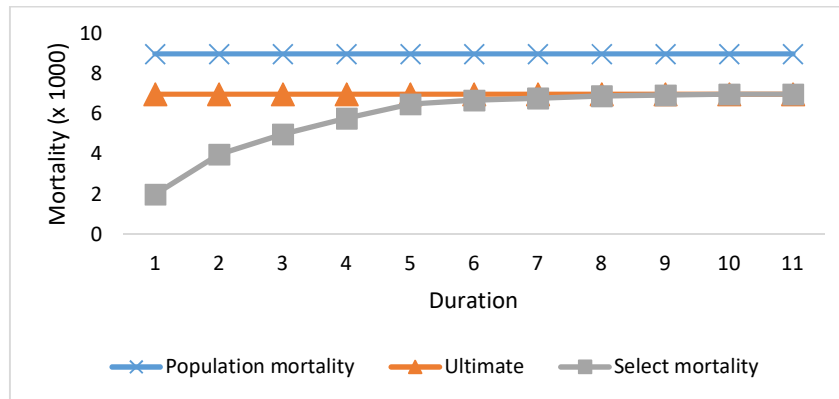
The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1: An “interest only” loan is a loan that is repayable by a series on interest payments followed by return of the initial loan amount.

A repayment or mortgage loan is a loan that is repayable by a series of payments that include partial repayment of the loan capital in addition to the interest payments

[2 Marks]

Solution 2: If we were to plot mortality for any given age x by varying duration and compare population mortality, ultimate mortality and select mortality, the graph would be:



[2]

Relationship between population mortality and ultimate mortality: Those taking out life insurance contracts tend to be more affluent in the society, who can afford to purchase insurance products. Financially better-off people tend to have lower mortality. Therefore, ultimate mortality is expected to be lower than population mortality.

[1.5]

Relationship between select mortality and ultimate mortality: Policyholders with duration of one year, say, will have recently satisfied the insurance company about their state of health. Therefore, we expect their mortality to be better than mortality of policyholders of the same age with longer duration (say 3 or 5 or 10 years). i.e. select mortality will be lower than ultimate mortality.

[1.5]

[3]

[5 Marks]

Solution 3:

i) Starting from LHS

$$(la)_{n1} = v + 2v^2 + 3v^3 + \dots + nv^n \quad (A) \quad [1]$$

Multiplying by $(1+i)$ both sides, we get

$$(1+i)(la)_{n1} = 1 + 2v + 3v^2 + \dots + nv^{n-1} \quad (B) \quad [0.5]$$

By subtraction $(B - A)$, we get:

$$\begin{aligned} i(la)_{n1} &= 1 + v + v^2 + v^3 + \dots + v^{n-1} - nv^n \\ &= \ddot{a}_{n1} - nv^n \end{aligned} \quad [0.5]$$

So:

$$(la)_{n1} = (\ddot{a}_{n1} - nv^n) / i \quad [2]$$

ii) Value of X :

The present value of contributions:

[0.5]

$$= 2,00,000 * \ddot{a}_{101} + X (la)_{91}$$

$$i = 10\%, v = 0.9091, d = i/(1+i) = 9.091\%$$

$$\ddot{a}_{101} = (1-v^{10})/d$$

$$= 6.7590$$

[0.5]

$$(la)_{91} = (\ddot{a}_{91} - 9v^9) / i$$

$$\text{Or, } \ddot{a}_{91} = (1-v^9)/d$$

$$= 6.3349$$

$$(la)_{91} = (6.3349 - 9 * .4241) / 0.1$$

$$= 25.1804$$

[1]

Thus present value of contribution is

$$= 2,00,000 * 6.759026 + X * 25.18046$$

$$= 13,51,805.20 + X * 25.18046$$

[0.5]

The present value of higher education expense:

$$= 30,00,000 * (1.06)^{10} * v^{10} @ 10\%$$

$$= 20,71,350$$

[0.5]

Substituting and equating the values:

$$20,71,350 = 13,51,805.20 + X * 25.18046$$

$$X = 28,575 \text{ approx}$$

[1]

[4]

[6 Marks]

Solution 4:

- i) Asset share: This is the amount of money that the company has accumulated from the policy, net of that policy's share of the company's expenses and the cost of death benefits that had been covered over the policy term.

[1]

Asset share is a tool used in setting bonuses. Upon claim, if asset share of a policy is in excess of the guaranteed benefit amount i.e. sum assured plus all accrued annual bonuses, then this excess can be distributed as terminal bonus.

[1]

[2]

- ii) Let the bonus rate be b% and Sum Assured be S.

If simple bonus rate applies, then bonus additions accrued by end of "n" policy years =

$$n * b\% * S$$

If compound bonus rate applies, then bonus additions accrued by end of "n" policy years =

$$S (1+b\%)^n - S$$

[0.5]

- Simple bonus applies only on sum assured, whereas compound applies on both Sum Assured as well as previously accrued bonus. [0.5]
- At the end of n years, total accrued bonus under simple bonus will be lower than if compound bonus had applied. [0.5]
- There will be larger deferral of distribution of surplus under simple bonus method. This would result in greater ability of company to smooth bonus rates, enable company to take greater risk which can result in overall higher returns to the policyholder [1]
- Therefore, where bonus is simple, terminal bonus may be higher [0.5]

- Where bonus is simple, premiums may be lower, as guaranteed benefits will grow at a slower rate and premium rates tend to include a loading for future bonuses.

[1]

[3]

[5 Marks]

Solution 5: The key steps in the data analysis process in this scenario are:

- Develop a well-defined set of objectives that need to be met by the results of the data analysis. Here the objective is to determine if there is a significant difference in probability of having an accident between a male and a female [0.5]
- Identify the data items required for the analysis. The data items required would include total number of male and female drivers and the number of accidents that they have had. [0.5]
- Collection of the data from appropriate sources. The insurer will have its own data during the investigation period and which of them had claimed. The insurer may also be able to buy data from an external source like Indian Insurance Bureau which collates information from various insurers. [1]
- Processing and formatting the data for analysis, e.g. inputting into a spreadsheet, database or other model. The data will need to be extracted from the IT or admin system and loaded into whichever statistical package is being used for the analysis. If different data sets are being combined, they will need to be put into a consistent format and any duplicates (the same records appearing in different data sets) will need to be removed. [0.5]
- Cleaning data, e.g. addressing unusual, missing or inconsistent values. For example, the gender of the driver might be missing [0.5]
- Exploratory data analysis, which here takes the form of inferential analysis as we are testing the hypothesis that there is a significant difference in probability of having an accident between male and female drivers. [0.5]
- Modelling the data. This may involve fitting a distribution to the annual number of accidents arising from the policyholders in both genders separately. [0.5]
- Communicating the results. This will involve describing the data sources used, the model and analyses performed, and the conclusion of the analysis (i.e. whether there is indeed any difference in likelihood to have an accident between the two genders), along with any limitations of the analysis. [1]
- Monitoring the process – updating the data and repeating the process if required. The car insurer may wish to repeat the process again in a few years' time, using the data gathered over that period, to ensure that the conclusions of the original analysis remain valid. [0.5]
- Ensuring that any relevant professional guidance and legislation (e.g. on gender discrimination) has been complied with. [0.5]

[6 Marks]

Solution 6:

- i) To calculate the reserve at time 1, we first need to calculate the annual premium for the policy. Let P be the level annual premium. The expected PV of the premium is:

$$P * \ddot{a}_{50:10} = 10,00,000 A^1_{50:10} \quad [0.5]$$

Using the tables:

$$\ddot{a}_{50:10} = 8.314 \text{ and} \quad [0.5]$$

$$A^1_{50:10} = (A_{50:10} - D_{60}/D_{50}) = 0.68024 - 882.85/1366.61 = 0.034225 \quad [0.5]$$

$$\text{Hence } P = \text{INR } 4,116.60 \quad [0.5]$$

The reserve at time 1 is:

$$\begin{aligned} {}_1V &= 10,00,000 * A^1_{51:9} - 4116.60 * \ddot{a}_{51:9} \\ &= 10,00,000 * (0.70672 - 882.85 / 1310.75) - 4116.60 * 7.652 \\ &= 1,674.10 \end{aligned} \quad [1]$$

The death strain at risk for each term assurance policy is then:

$$\text{DSAR} = S - {}_1V = 10,00,000 - 1,674.10 = \text{INR } 998,325.9 \quad [1]$$

[4]

- ii) There are $2000 - 3 = 1997$ active policies on 1st Jan, 2018 [0.5]

The expected death strain for this group of policies would be:

$$\begin{aligned} \text{EDS} &= 1997 * q_{51} * 998,325.9 \\ &= 1997 * 0.002809 * 998,325.9 \\ &= 5,600,182 \end{aligned} \quad [1]$$

The actual death strain for this group of policies is:

$$\text{ADS} = 4 * 998,325.9 = 3,993,303.6 \quad [0.5]$$

So the mortality profit from this group of policies is:

$$\text{MP} = \text{EDS} - \text{ADS} = \text{INR } 1,606,878.4 \quad [1]$$

[3]

[7 Marks]

Solution 7:

We will use notations as ${}_t(ap)_x'$ and ${}_t(aq)_x'$ to represent dependent probabilities that *ignore*

decrement *retirement*. We can then write:

$$\begin{aligned} (aq)_{59.5}^{(\text{attrition})} &= 0.5 (aq)_{59.5}^{(\text{attrition})'} + 0.5 (ap)_{59.5}' * (1 - q_{60}^{(\text{retire})}) * 0.5 \\ (aq)_{60}^{(\text{attrition})'} \end{aligned}$$

In order for the life to leave the population due to decrement a in the year of age (59.5, 60.5), it could either leave due to decrement attrition in the first half year, or, if it remains in the workforce until age 60, and does not leave due to decrement retirement at that age, it could leave due to decrement attrition again in the last half of the year. [1]

Now for $0 \leq t \leq 1$ and $0 \leq t \leq 1$ t :

$$\begin{aligned} {}_t(ap)_{x+r}' &= \exp[-(\text{decrement_attrition} + \text{decrement_mortality}) * t] \\ &= \exp(-0.025t) \text{ and,} \\ {}_t(aq)_{x+r}^{(\text{attrition})'} &= \text{decrement_attrition} / (\text{decrement_attrition} + \text{decrement_mortality}) * (1 - {}_t(ap)_{x+r}') \\ &= 0.02/0.025 * (1 - \exp(-0.025t)) \\ {}_t(aq)_{x+r}^{(\text{death})'} &= \text{decrement_death} / (\text{decrement_attrition} + \text{decrement_mortality}) * (1 - {}_t(ap)_{x+r}') \\ &= 0.005/0.025 * (1 - \exp(-0.025t)) \end{aligned} \quad [1]$$

Hence

$$\begin{aligned} {}_{0.5}(aq)_{59.5}^{(\text{attrition})'} &= 0.8 * (1 - \exp(-.025 * .5)) \\ &= 0.8 * 0.012422 \\ &= 0.009938 \end{aligned}$$

Also, ${}_{0.5}(aq)_{60}^{(\text{attrition})'}$ would be same as ${}_{0.5}(aq)_{59.5}^{(\text{attrition})'}$ because rate of attrition is constant over the year. [1]

$$\begin{aligned} {}_{0.5}(aq)_{59.5}^{(\text{death})'} &= {}_{0.5}(aq)_{60}^{(\text{death})'} \\ &= 0.2 * 0.012422 \\ &= 0.002484 \end{aligned}$$

[1]

$$\begin{aligned} {}_{0.5}(ap)_{59.5}' &= \exp(-.025 * .5) \\ &= 0.98758 \end{aligned}$$

[1]

Therefore,

$$\begin{aligned} (aq)_{59.5}^{(\text{attrition})} &= 0.009938 + 0.98758 * (1 - 0.4) * 0.009938 \\ &= 0.015827 \end{aligned}$$

[1]

Similarly, for death:

$$\begin{aligned} (aq)_{59.5}^{(\text{death})} &= 0.002484 + 0.98758 * (1 - 0.4) * 0.002484 \\ &= 0.003956 \end{aligned}$$

[1]

For retirement:

$$\begin{aligned} (aq)_{60}^{(\text{retire})} &= {}_{0.5}(ap)_{59.5}' * q_{60}^{(\text{retire})} \\ &= 0.98758 * 0.4 \\ &= 0.39503 \end{aligned}$$

[0.5]

Number of employees leaving due to

Attrition = $1000 * 0.015827 = 15.83$ rounded to 16

Death = $1000 * 0.003956 = 3.956$ rounded to 4

Retirement = $1000 * 0.39503 = 395.03$ rounded to 395

[1.5]

[9 Marks]

Solution 8:

i)

The price P of the perpetual bond is given by the following formula:

$$\begin{aligned} P &= \sum_{y=1}^{\infty} \frac{N * C}{(1+i)^y} \\ &= \frac{N * C}{i} \end{aligned}$$

[1]

$$\begin{aligned} \text{The duration D is: } &-(1+i) \frac{P'(i)}{P(i)} \\ &= -(1+i) \frac{-C/i^2}{C/i} \\ &= \frac{1+i}{i} \end{aligned}$$

[1]

[2]

a) For duration neutral strategy:

$$D_P = \sum_{i=1}^n w_i D_i = 0$$

Let i be the semi-annual rate of interest,

Therefore i = 4% per half year

For Bond A:

Price

Let nominal price be 100 for both A and B

$$\text{Price (A)} = 4.75 a_{361}^{@i} + 100 v^{36@i}$$

$$a_{361}^{@i} = 18.90828$$

$$v^{36} = 0.243669$$

$$\text{Price (A)} = 114.181$$

[1]

Duration

$$\frac{\sum t C_t v^t}{\sum C_t v^t}$$

Working in half-years

$$\text{Numerator} = 4.75 (Ia)_{36|} + 36 \cdot 100 \cdot v^{36|i} \quad [1]$$

$$= 4.75 \cdot 272.3135 + 3600 \cdot 0.243669$$

$$= 2,170.69$$

$$\text{Denominator} = \text{Price (A)} = 114.181$$

$$\text{Duration (A)} = 2170.698 / 114.181$$

$$= 19.011 \text{ half -years}$$

$$= 9.5055 \text{ years} \quad [1]$$

[3]**b) Similarly, for Bond B****Price**

$$\text{Price (B)} = 5 a_{40|}^{@i} + 100 v^{40|i}$$

$$a_{40|}^{@i} = 19.79277$$

$$v^{40} = 0.208289$$

$$\text{Price (B)} = 119.792 \quad [1]$$

Duration

Working in half-years

Numerator:

$$= 5 (Ia)_{40|} + 40 \cdot 100 \cdot v^{40|i}$$

$$= 5 \cdot 306.32 + 4000 \cdot 0.2082$$

$$= 2364.77 \quad [1]$$

$$\text{Denominator} = \text{Price (B)} = 119.792$$

$$\text{Duration (B)} = 2364.772 / 119.792$$

$$= 19.74 \text{ half -years}$$

$$= 9.87 \text{ years} \quad [1]$$

[3]**c)**Let ϕ be the number of units of Bond B

Therefore, as sum of weighted average of duration of bonds is zero,

$$100,000 \cdot 114.181 \cdot 9.5055 + \phi \cdot 119.792 \cdot 9.8703 = 0 \quad [1]$$

$$\text{i.e. } \phi = - \frac{11,418,100 \cdot 9.5055}{119.792 \cdot 9.8703} = - 91,793$$

The trader needs to sell 91,793 units of the Bond B. [1]

[2]**[10 Marks]****Solution 9:****i)** The independent probabilities of mortality are:

$$q_{50}^d = q_{[50]} = 0.001971$$

$$q_{51}^d = q_{[50]+1} = 0.002732$$

$$q_{52}^d = q_{52} = 0.003152$$

$$q_{53}^d = q_{53} = 0.003539 \quad [0.5]$$

So we have:

$$p_{50}^d = 0.998029 \text{ and } p_{50}^w = 0.93$$

$$p_{51}^d = 0.997268 \text{ and } p_{51}^w = 0.93$$

$$p_{52}^d = 0.996848 \text{ and } p_{52}^w = 0.93$$

$$p_{53}^d = 0.996461 \text{ and } p_{50}^w = 0.99 \quad [0.5]$$

Hence,

$$(ap)_{50} = p_{50}^d * p_{50}^w = 0.998029 * 0.93 = 0.928167$$

$$(ap)_{51} = p_{51}^d * p_{51}^w = 0.997268 * 0.93 = 0.927459$$

$$(ap)_{52} = p_{52}^d * p_{52}^w = 0.996848 * 0.93 = 0.927069$$

$$(ap)_{53} = p_{53}^d * p_{53}^w = 0.996461 * 0.99 = 0.986496 \quad [1]$$

We do not need reserve at time 4 since the cashflow at time 5 is positive.

$$\text{Hence } {}_4V = 0 \quad [0.5]$$

We do need a reserve at time 3 since the cashflow at time 4 is negative.

We require:

$${}_3V(1+i) = 40 \text{ where } i = 8\%$$

$$\text{Hence } {}_3V = 37.037 \quad [0.5]$$

The non-unit cashflow at time 3 then becomes:

$$(NUCF)'_3 = 30 - {}_3V * (ap)_{52}$$

$$= 30 - 37.037 * 0.9271$$

$$= 30 - 34.3358$$

$$= -4.336 \quad [0.5]$$

We now need a reserve at time 2 to zeroise this negative:

$${}_2V(1+i) = 4.336, \text{ where } i = 8\%$$

$${}_2V = 4.015 \quad [0.5]$$

The non-unit cashflow at time 2 then becomes:

$$(NUCF)'_2 = 10 - {}_2V * (ap)_{51}$$

$$= 10 - 4.0146 * 0.927459$$

$$= 10 - 3.723$$

$$= 6.277 \quad [0.5]$$

Since this is positive, we do not need a reserve at time 1, hence ${}_1V = 0$ [0.5]

[5]

- ii) In order to find the net present value of profit, we need the probabilities of survival:

$${}_2(ap)_{50} = (ap)_{50} * (ap)_{51} = 0.860837$$

$${}_3(ap)_{50} = {}_2(ap)_{50} * (ap)_{52} = 0.798055$$

$${}_4(ap)_{50} = {}_3(ap)_{50} * (ap)_{53} = 0.787278 \quad [1]$$

So the net present value of the contract would be:

$$= -100/1.09 + 10/1.09^2 * 0.928167 + 30/1.09^3 * 0.860837 -$$

$$40/1.09^4 * 0.798055 + 210/1.09^5 * 0.787278$$

$$= 20.85 \quad [1]$$

After zeroisation, the profit vector becomes (-100, 6.27659, 0, 0, 210) [1]

Net present value would be:

$$= -100/1.09 + 6.27659/1.09^2 * 0.928167 + 210/1.09^5 * 0.787278$$

$$= 20.61 \quad [1]$$

As expected, the NPV after zeroisation is smaller because losses are realised earlier and the risk discount rate (9%) is greater than the accumulation rate (8%). [1]

[5]

- iii) Holding reserves delays the emergence of profit. However, if the rate of interest earned on the reserves is 9%, then we are accumulating and discounting at the same rate. In this case, delaying the emergence of profit will have no effect on the net present value for the contract.
So the net present value would be 20.85.

[1]

[11 Marks]

Solution 10:

- a. Prepayment at the end of 2 years:

Bank A:

Interest rate $i^{(12)} = 9\%$ p.a.

Let i be the effective monthly rate of interest

$$i = 9\% / 12 = 0.75\%$$

Let monthly instalment be X

$$10,00,000 = X a_{961} @ i$$

$$X = 14,650.20 \text{ per month.}$$

[1]

Outstanding loan at the end of 2 years

$$= 14,650.20 * a_{721} @ i$$

$$= 14,650.20 * 55.4768$$

$$= 8,12,746.93$$

[1]

Total instalments paid in 2 years = $24 * 14,650.20 = 3,51,604.80$

Interest paid in 2 years = $3,51,604.80 - (10,00,000 - 8,12,746.93)$

$$= 1,64,351.73$$

Total interest payable under the loan arrangement =

$$\text{Loan Term} * 12 * \text{instalment} - \text{loan amount}$$

$$= 8 * 12 * 14650.20 - 10,00,000$$

$$= 4,06,419.20$$

Unpaid interest = $4,06,419.20 - 1,64,351.73 = 2,42,067.47$

[1]

Prepayment charges = MIN (5% of outstanding loan , unpaid interest)

$$= \text{MIN}(5\% * 8,12,746.93, 2,42,067.47)$$

$$= 40,637.35$$

[0.5]

Total payment in 2 years if loan is repaid in full:

= Processing fee + loan outstanding to be repaid + total instalments paid in

2 years + prepayment charges

$$= 3\% * 10,00,000 + 8,12,746.93 + 3,51,604.80 + 40,637.35$$

$$= 12,34,989$$

Additional amount paid over loan = $12,34,989 - 10,00,000$

$$= 2,34,989$$

[1]

Bank B:

Interest rate $i^{(12)} = 10.5\%$ p.a.

Let i be the effective monthly rate of interest

$$i = 10.5\% / 12 = 0.875\%$$

Let monthly instalment be X

$$10,00,000 = X a_{961} @ i$$

$$X = 15,440.02 \text{ per month.}$$

[1]

Outstanding loan at the end of 2 years

$$= 15,440.02 * a_{721} @ i$$

$$= 15,440.02 * 53.25$$

$$= 8,22,197.38$$

[1]

Total instalments paid in 2 years

$$= 24 * 15,440.02 = 3,70,560.50$$

Prepayment charges = 2% of outstanding loan

$$= 2\% * 8,22,197.38$$

$$= 16,443.95$$

[0.5]

Total payment in 2 years if loan is repaid in full:

= loan outstanding to be repaid + total instalments paid in 2 years + prepayment charges

$$= 0\% * 10,00,000 + 8,22,197.38 + 3,70,560.50 + 16,443.95$$

$$= 12,09,201$$

$$\text{Additional amount paid over loan} = 12,09,201 - 10,00,000$$

$$= 2,09,201$$

Hence under option (i) Bank B should be chosen as the amount paid till repayment at the end of 2 years is less as compared to Bank A

[1]

b. Prepayment at the end of 7 years

Bank A:

Outstanding loan at the end of 7 years

$$= 14,650.20 * a_{12}^{i@}$$

$$= 14,650.20 * 11.44$$

$$= 1,67,523.76$$

[1]

Total instalments paid in 7 years = $84 * 14,650.20 = 12,30,616.80$

Interest paid in 7 years = $12,30,616.80 - (10,00,000 - 1,67,523.76)$

$$= 3,98,140.56$$

Total interest payable under the loan arrangement = Loan Term * 12 *

instalment – loan amount = $8 * 12 * 14,650.20 - 10,00,000$

$$= 4,06,419.20$$

Unpaid interest = $4,06,419.20 - 3,98,140.56$

$$= 8,278.64$$

[1]

Prepayment charges = MIN (5% of outstanding loan , unpaid interest)

$$= \text{MIN} (5\% * 1,67,523.76, 8,278.64)$$

$$= 8,278.64$$

[0.5]

Total payment in 7 years if loan is repaid in full:

= Processing fee + loan outstanding to be repaid + total instalments paid in 7 years + prepayment charges

$$= 3\% * 10,00,000 + 1,67,523.76 + 12,30,616.80 + 8,278.64$$

$$= 14,36,419$$

$$\text{Additional amount paid over loan} = 14,36,419 - 10,00,000 = 4,36,419$$

[1]

Bank B:

Outstanding loan at the end of 7 years

$$= 15,440.02 * a_{12}^{i@}$$

$$= 15,440.02 * 11.344479$$

$$= 1,75,158.98$$

[1]

Total instalments paid in 7 years

$$= 84 * 15,440.02$$

$$= 12,96,961.68$$

Prepayment charges = 0 as prepayment is after 3 years

[0.5]

Total payment in 7 years if loan is repaid in full:

$$\begin{aligned}
 &= \text{loan outstanding to be repaid} + \text{total instalments paid in 7 years} \\
 &= 1,75,158.98 + 12,96,961.68 \\
 &= 14,72,121
 \end{aligned}$$

Additional amount paid over loan = 14,72,121 – 10,00,000

$$= 4,72,121$$

Hence under option (ii) Bank A should be chosen as the amount paid till repayment at the end of 7 years is less as compared to Bank A

[1]

[14 Marks]

Solution 11:

i)

SA = 50,00,000, decreasing at 500,000 payable at end of year of death

Gross premium = P

Policy Term, N = 10 years

Premium Payment term, PPT = 5 years; with premiums being paid quarterly in advance

Renewal expense = e = 500 p.a. from start of year 2 to start of year 10

Initial expense = f = 0.2 * P

Renewal commission = 5% * P, payable quarterly in advance from second policy year along with the premium

Initial commission = 10% * P

For the pricing basis, mortality = 100% AM 92 and interest rate = 6%.

By principle of equivalence, we have that:

PV Premium = PV Benefits + PV Expenses + PV Commissions

PV Benefits = 50,00,000 ((10 + 1) $A^1_{30:10}$ – $IA^1_{30:10}$)

[1]

$$A^1_{30:10} = A_{30} - l_{40} / l_{30} v^{10} A_{40}$$

$$= 0.07328 - 9856.2863 / 9925.20194 * 1.06^{-10} * 0.12313$$

$$= 0.07328 - 0.068278$$

$$= 0.0050$$

[1]

$$IA^1_{30:10} = IA_{30} - l_{40} / l_{30} v^{10} (IA_{40} + 10 * A_{40})$$

$$= 2.84692 - 9856.2863 / 9925.2094 * 1.06^{-10} * (3.85435 + 10 * 0.12313)$$

$$= 2.84692 - 2.82008$$

$$= 0.02684$$

[1]

$$\text{PV Benefits} = 500,000 * (11 * 0.0050 - 0.02684) = \mathbf{14,093}$$

[0.5]

$$\text{PV Expenses} = 0.2 * P + 500 a_{30:9}$$

[0.5]

$$= 0.2P + 500 (a_{30} - v^9 * l_{39} / l_{30} * a_{39})$$

$$= 0.2P + 500 (15.372 - 1.06^{-9} * 9864.8680 / 9925.2094 * 14.602)$$

$$= 0.2P + 3390.82$$

[1]

$$\text{PV Commissions} = 0.1P + 5\% P * v * l_{31} / l_{30} * \ddot{a}^{(4)}_{31:4}$$

[1]

$$\ddot{a}^{(4)}_{31:4} = \ddot{a}^{(4)}_{31} - l_{35} / l_{31} * v^4 \ddot{a}^{(4)}_{35}$$

$$\ddot{a}^{(4)}_{31} = \ddot{a}_{31} - 3 / 8 = 15.929$$

$$\ddot{a}^{(4)}_{35} = \ddot{a}_{35} - 3 / 8 = 15.615$$

$$\text{PV Commission} = 0.1P + 0.05P / 1.06 * 9919.3535 / 9925.2094 * 3.591542$$

$$= 0.1148 P$$

[1]

$$PV \text{ Premium} = P \ddot{a}^{(4)}_{30:5}$$

$$\ddot{a}^{(4)}_{30:5} = \ddot{a}^{(4)}_{30} - l_{35}/l_{30} * v^5 \ddot{a}^{(4)}_{35}$$

$$\ddot{a}^{(4)}_{30} = \ddot{a}_{30} - 3 / 8 = 15.997$$

$$\ddot{a}^{(4)}_{35} = 15.615$$

$$PV \text{ Premium} = 4.3647 P$$

[1]

By principle of equivalence, we have:

$$4.364749P = 0.114766P + 14,093 + 0.2P + 3390.822$$

$$P = 4317.01$$

$$\text{Gross premium} = 4317.01$$

[1]

[9]

ii)

Profit in the fifth policy year under the gross premium method =

$$250 * [(4V + 0.95 P * \ddot{a}^{(4)}_{34:1} @ 6\% - 500) * 1.06 - 80\% * q_{34} * 30,00,000 - (1 - 80\% * q_{34})_5 V]$$

[2]

To calculate GPV at the end of year 4 and end of year 5. We now use mortality = 100% AM 92 and interest as 4%.

Calculation of benefits:

PV Benefits at the end of year 4:

$$= 35,00,000 A^1_{34:6} - 500,000 IA^1_{34:6}$$

[1]

$$A^1_{34:6} = A_{34} - l_{40}/l_{34} v^6 A_{40}$$

$$= 0.18531 - 9856.2863 / 9900.9645 * 1.04^{-6} * 0.23056$$

$$= 0.00392$$

$$IA^1_{34:6} = IA_{34} - l_{40}/l_{34} v^6 (IA_{40} + 6 * A_{40})$$

$$= 7.36239 - 9856.2863 / 9900.9645 * 1.04^{-6} * (7.9570 + 6 * 0.23056)$$

$$= 0.01389$$

[1]

$$PV \text{ Benefits at the end of year 4} = 6,766.39$$

[0.5]

PV Benefits at the end of year 5:

$$= 30,00,000 A^1_{35:5} - 500,000 IA^1_{35:5}$$

[1]

$$A^1_{35:5} = A_{35} - l_{40}/l_{35} v^5 A_{40}$$

$$= 0.19219 - 9856.2863 / 9894.4299 * 1.04^{-5} * 0.23056$$

$$= 0.00342$$

$$IA^1_{35:5} = IA_{35} - l_{40}/l_{35} v^5 (IA_{40} + 5 * A_{40})$$

$$= 7.46909 - 9856.2863 / 9894.4299 * 1.04^{-5} * (7.95699 + 5 * 0.23056)$$

$$= 0.010372$$

[1]

$$PV \text{ Benefits at the end of year 5} = 5,065.24$$

[0.5]

Calculation of PV Premium:

$$PV \text{ Premium at the end of year 4} = P \ddot{a}^{(4)}_{34:1}$$

$$\ddot{a}^{(4)}_{34:1} = \ddot{a}^{(4)}_{34} - l_{35}/l_{34} * v \ddot{a}^{(4)}_{35}$$

$$\ddot{a}^{(4)}_{34} = \ddot{a}_{34} - 3 / 8 = 20.807$$

$$\ddot{a}^{(4)}_{35} = \ddot{a}_{35} - 3 / 8 = 20.628$$

$$\text{So, } \ddot{a}^{(4)}_{34:1} = 0.985475$$

$$PV \text{ Premium at the end of year 4} = 4317 * 0.985475 = 4,254.30$$

[1.5]

$$PV \text{ Premium at the end of year 5} = 0$$

[0.5]

Calculation of PV commission:

$$PV \text{ commission at the end of year 4} = 5\% * 4,254.297 = 212.72$$

$$PV \text{ Commission at the end of year 5} = 0$$

[1]

Calculation of PV renewal expenses:

PV of renewal expenses at the end of year 4 = $500 \ddot{a}_{34:6}$

$$\begin{aligned}\ddot{a}_{34:6} &= \ddot{a}_{34}^{(4)} - l_{40}/l_{34} * v^6 \ddot{a}_{40} \\ &= 21.182 - 9856.2863 / 9900.9645 * 1.04^{-6} * 20.005 \\ &= 5.443102\end{aligned}$$

So, PV expenses at the end of year 4 = $500 * 5.443102 = \mathbf{2,721.55}$ [1]

PV of renewal expenses at the end of year 5 = $500 \ddot{a}_{35:5}$

$$\begin{aligned}\ddot{a}_{35:5} &= \ddot{a}_{35}^{(4)} - l_{40}/l_{35} * v^5 \ddot{a}_{35} \\ &= 21.003 - 9856.2863 / 9894.429 * 1.04^{-5} * 20.005 \\ &= 4.623734\end{aligned}$$

So, PV expenses at the end of year 5 = $500 * 4.623734 = \mathbf{2311.87}$ [1.5]

Reserve at any point of time = PV Benefits – PV Expenses – PV

Commissions – PV Premiums

So,

Reserve at the end of year 4 = $\mathbf{5,446.35}$ [0.5]

Reserve at the end of year 5 = $\mathbf{7,377.10}$ [0.5]

Profit in the fifth policy year under the gross premium method =

$$250 * [(4V + 0.95 P * \ddot{a}_{34:1}^{(4)} @ 6\% - 500) * 1.06 - 80\% * q_{34} * 30,00,000 - (1 - 80\% * q_{34}) {}_5V]$$

$$\ddot{a}_{34:1}^{(4)} = \ddot{a}_{34}^{(4)} - l_{35}/l_{34} * v \ddot{a}_{35}^{(4)}$$

$$\ddot{a}_{34}^{(4)} = 16.075 - 3 / 8$$

$$\ddot{a}_{35}^{(4)} = 15.990 - 3 / 8$$

So, $\ddot{a}_{34:1}^{(4)} = 0.97859$ [1]

Therefore, profit =

$$\begin{aligned}250 * [(5446.354 + 0.95 * 4317.01 * 0.97859 - 500) * 1.06 - 80\% * \\ 0.000660 * 30,00,000 - (1 - 80\% * 0.000660) * 7377.102] \\ = 250 * 540.08 \\ = \mathbf{135,021}\end{aligned}$$

[1.5]

[16]

[25 Marks]
